

Piecewise Self-Similar Radiation-Hydrodynamics Analytic Profiles

Antigravity & Tal Ramot

May 2026

1 Wave Front and Mass Coordinates

All physical quantities are scaled using nanoseconds (ns) for time, g/cm^2 for mass coordinates, g/cm^3 for density, MBar for pressure, km/s for velocity, and HeV for temperature.

Exact physical constants (according to SI):

Quantity	value	SI units
c	299792458	m / s
k_B	1.380649×10^{-23}	J / K
q	$1.602176634 \times 10^{-19}$	J
h	$6.62607015 \times 10^{-34}$	J s

Other fundamental quantities derived are:

Quantity	Theoretical Expression (SI)	CGS Units
HeV	$\frac{q}{k_B} \cdot 100$	K
σ_{SB}	$\frac{2\pi^5 k_B^4}{15c^2 h^3}$	$\text{erg}/(\text{cm}^2 \cdot \text{s} \cdot \text{K}^4)$
a	$\frac{4\sigma_{SB}}{c}$	$\text{erg}/(\text{cm}^3 \cdot \text{K}^4)$

Case 1: $\tau = 0$

Material model.

Quantity	value	CGS units
f	3.4×10^{13}	erg / g
f_{Kelvin}	$\frac{f}{\text{HeV}^\beta}$	erg / g
β	1.6	-
μ	-0.14	
g	7200	cm^2 / g
g_{Kelvin}	$\frac{g}{\text{HeV}^\alpha}$	erg / g
α	1.5	-
λ	0.2	-
γ	1.25	-
r	0.25	-
ρ_0	19.32	g/cm^3

1.1 Self Similar Constants

$$A = T_0(rf)^{\frac{1}{\beta}} = \dots \quad (1)$$

$$B = \frac{16\sigma g}{3\beta(rf)^{\frac{4+\alpha}{\beta}}} = \dots \quad (2)$$

$$n = \frac{4+\alpha}{\beta} = \frac{5.5}{1.6} = \frac{55}{16} \quad (3)$$

$$k = 1 - \mu = 1 - 0.14 = 0.86 \quad (4)$$

$$a_1 = \frac{\mu}{\beta} = \frac{7}{80} \quad (5)$$

$$a_2 = \frac{2-\mu}{\beta} = \frac{93}{80} \quad (6)$$

$$a_3 = -\frac{2}{\beta} - \tau = -\frac{5}{4} \quad (7)$$

$$b_1 = \frac{\beta(2+\lambda) - \mu(4+\alpha)}{\beta} = \frac{55}{32} \quad (8)$$

$$b_2 = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{\beta} = -\frac{723}{160} \quad (9)$$

$$b_3 = \frac{8 - 3\beta + 2\alpha}{\beta} = \frac{31}{8} \quad (10)$$

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{4 + 2\lambda - 4\mu} = -\frac{723}{412} \quad (11)$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} = -\frac{93}{206} \quad (12)$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau(2(\beta - \alpha - 4) - \beta\lambda + \mu(4+\alpha))}{4 + 2\lambda - 4\mu} = -\frac{89}{206} \quad (13)$$

$$a_v = \frac{2\beta - 4 - \alpha}{2 + \lambda - 2\mu} = -\frac{115}{96} \quad (14)$$

$$b_v = \frac{-1}{2 + \lambda - 2\mu} = -\frac{25}{48} \quad (15)$$

$$c_v = \frac{1 - \tau(4 + \alpha - 2\beta)}{2 + \lambda - 2\mu} = \frac{25}{48} \quad (16)$$

$$a_u = \frac{\beta(2+\lambda) - \mu(4+\alpha)}{4 + 2\lambda - 4\mu} = \frac{275}{412} \quad (17)$$

$$b_u = \frac{-\mu}{4 + 2\lambda - 4\mu} = -\frac{7}{206} \quad (18)$$

$$c_u = \frac{\tau(\beta(2+\lambda) - \mu(4+\alpha)) + \mu}{4+2\lambda-4\mu} = \frac{7}{206} \quad (19)$$

$$a_p = \frac{\beta\lambda + 4 + \alpha - \mu(4+\alpha)}{2+\lambda-2\mu} = \frac{505}{192} \quad (20)$$

$$b_p = \frac{1-\mu}{2+\lambda-2\mu} = \frac{43}{96} \quad (21)$$

$$c_p = \frac{\tau(\beta\lambda + 4 + \alpha - \mu(4+\alpha)) + \mu - 1}{2+\lambda-2\mu} = -\frac{43}{96} \quad (22)$$

$$\begin{aligned} v(m, t) &= V(\xi) A^{a_v} B^{b_v} t^{c_v} = V(\xi) A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}} \\ &\approx 5.749439 \left(1 - \frac{m}{m_f(t)}\right)^{0.491367} A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}} \\ &\approx 1.180662 \times 10^{-4} \left(1 - 981.4602 m \left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}\right)^{0.491367} A^{-\frac{115}{96}} B^{-\frac{25}{48}} \left(\frac{t}{\text{ns}}\right)^{\frac{25}{48}} \end{aligned} \quad (23)$$

$$\rho(m, t) = \left(\frac{rf T(m, t)^\beta}{p(m, t)}\right)^{\frac{1}{\mu-1}} \quad (24)$$

Note: The density profile diverges sharply at the ablation front ($\xi \rightarrow 1$), so instead of independently fitting $\tilde{\rho}(\xi)$, we compute it explicitly using the equation of state (EOS) along with the well-behaved analytical fits for pressure $p(m, t)$ and temperature $T(m, t)$.

$$\begin{aligned} u(m, t) &= U(\xi) A^{a_u} B^{b_u} t^{c_u} = U(\xi) A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}} \\ &\approx -6.46229 \frac{1 - \frac{m}{m_f(t)}}{1 + 4.78201 \frac{m}{m_f(t)}} A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}} \\ &\approx -3.03570 \frac{1 - 981.4602 m \left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}{1 + 4693.35 m \left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}} A^{\frac{275}{384}} B^{-\frac{7}{192}} \left(\frac{t}{\text{ns}}\right)^{\frac{7}{192}} \end{aligned} \quad (25)$$

$$\begin{aligned} p(m, t) &= P(\xi) A^{a_p} B^{b_p} t^{c_p} = P(\xi) A^{\frac{505}{192}} B^{\frac{43}{96}} t^{-\frac{43}{96}} \\ &\approx 7.051327 \left[0.348591 \left(\frac{m}{m_f(t)}\right)^{0.876766} + 0.029050 \left(\frac{m}{m_f(t)}\right)^{20.948361} \right] A^{\frac{505}{192}} B^{\frac{43}{96}} t^{-\frac{43}{96}} \\ &\approx 7.577405 \times 10^4 \left[0.348591 \left(981.4602 m \left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}\right)^{0.876766} \right. \\ &\quad \left. + 0.029050 \left(981.4602 m \left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}\right)^{20.948361} \right] A^{\frac{505}{192}} B^{\frac{43}{96}} \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}} \end{aligned} \quad (26)$$

$$e(m, t) = \frac{P(\xi)V(\xi)}{r} A^{\left(\frac{505}{192} - \frac{115}{96}\right)} B^{\left(\frac{43}{96} - \frac{25}{48}\right)} t^{\left(-\frac{43}{96} + \frac{25}{48}\right)} \quad (27)$$

$$= \frac{P(\xi)V(\xi)}{r} A^{\frac{275}{192}} B^{-\frac{7}{96}} t^{\frac{7}{96}} \quad (28)$$

$$T(m, t) = T(\xi)T_0t^0 = T(\xi)T_0 \quad (29)$$

1.2 Ablation (Heat) Front

The ablation front in similarity coordinates is $\xi_f \approx 0.427735$. In physical coordinates, the total ablated mass grows as a power law in time:

$$m_f(t) = xsi_f A^{-a} B^{-b} t^{-c} = 0.427735 \cdot m_{f0} \left(\frac{t}{\text{ns}}\right)^{0.515625} \approx 0.00767554 \left(\frac{t}{\text{ns}}\right)^{0.515625} \text{ g/cm}^2 \quad (30)$$

1.3 Shock Front

The shock front mass in similarity coordinates is $\xi_s \approx 1.669247$. The total shocked mass scales as:

$$m_s(t) = m_{s0} \left(\frac{t}{\text{ns}}\right)^{0.7765} \approx 0.0120786 \left(\frac{t}{\text{ns}}\right)^{0.7765} \text{ g/cm}^2 \quad (31)$$

2 Explicit Piecewise Profiles

2.1 Subsonic Ablation Region ($m < m_f(t)$)

In this region, the similarity variable is normalized to:

$$y = \frac{m}{m_f(t)} \in [0, 1] \quad (32)$$

Density Profile $\rho(m, t)$: The bulk density profile is derived algebraically from the temperature and pressure fits via the physical EOS:

$$\rho(m, t) = \left(\frac{r f_{\text{Kelvin}} T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}} \quad (33)$$

Pressure Profile $p(m, t)$: The pressure scales as:

$$p(m, t) \approx 7.05133 \left(\frac{t}{\text{ns}}\right)^{-0.44792} [0.34866y^{0.87714} + 0.02903y^{21.08862}] \text{ MBar} \quad (34)$$

Velocity Profile $u(m, t)$: We provide two alternative high-accuracy fits for the velocity profile:

- **Rational Fit** ($R^2 = 0.9689$):

$$u(m, t) \approx -191.294 \left(\frac{t}{\text{ns}}\right)^{0.03646} \frac{1-y}{1+4.78201y} \text{ km/s} \quad (35)$$

- **Rational Fractional Fit** ($R^2 = 0.9849$):

$$u(m, t) \approx -289.838 \left(\frac{t}{\text{ns}}\right)^{0.036458} \frac{1-y^{0.2340}}{1+0.7524y} \text{ km/s} \quad (36)$$

Temperature Profile $T(m, t)$: The temperature profile satisfies:

$$T(m, t) \approx [(1 - y)(1 + 0.20224 y)]^{\frac{10}{39}} \text{ HeV} \quad (37)$$

Note that at the boundary $m = 0$, $T(0, t) \approx 1 \text{ HeV}$ is constant.

2.2 Shock Region ($m_f(t) < m < m_s(t)$)

In the shocked region, the similarity variable is normalized to:

$$y_s = \frac{m - m_f(t)}{m_s(t) - m_f(t)} \in [0, 1] \quad (38)$$

Density Profile $\rho(m, t)$: Rather than using an independent curve fit, the density in the shocked region is algebraically derived from the pressure and temperature fits via the physical equation of state (EOS):

$$\rho(m, t) = \left(\frac{r f_{\text{Kelvin}} T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}} \quad (39)$$

At the shock front $y_s = 1$, the density reaches the strong shock Hugoniot limit of $\rho_s = 9\rho_0 = 173.88 \text{ g/cm}^3$.

Pressure Profile $p(m, t)$: The pressure profile scales as:

$$p(m, t) \approx 2.710000 \left(\frac{t}{\text{ns}} \right)^{-0.447} [1.0 + 0.493384 y_s^{1.133029}] \text{ MBar} \quad (40)$$

At the piston boundary $y_s = 0$, the pressure is precisely matched to the subsonic ablation pressure $p(m_f, t) \approx 2.71 (t/\text{ns})^{-0.447} \text{ MBar}$.

Velocity Profile $u(m, t)$: The velocity satisfies:

$$u(m, t) \approx \left(\frac{t}{\text{ns}} \right)^{-0.2235} [3.65776 + 0.65734 y_s^{0.644937}] \text{ km/s} \quad (41)$$

At the piston boundary $y_s = 0$, $u(m_f, t) \approx 3.65776 (t/\text{ns})^{-0.2235} \text{ km/s}$.

Temperature Profile $T(m, t)$: Using the ideal gas relation $T = p/((\gamma - 1)\rho)$ in HeV:

$$T(m, t) \approx 3.2091 \times 10^5 \left(\frac{t}{\text{ns}} \right)^{-0.447} y_s^{-0.438483} \text{ HeV} \quad (42)$$

Note the temperature singularity at the piston boundary $y_s \rightarrow 0$ due to the vanishing density.